INCH POUND

MIL-B-0044266A(GL)
28 October 1988
USED IN LIEU OF
MIL-T-44266
31 December 1986

MILITARY SPECIFICATION

BODY ARMOR, TROUSERS, EXPLOSIVE ORDNANCE DISPOSAL (EOD) SYSTEM

This specification is approved for use by the U.S. Army Natick Research, Development, and Engineering Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-B-44266. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers body armor trousers for protection against high velocity fragments associated with Unexploded Ordnance (UXO) and Improvised Explosive Device (IED). The insert is of aramid ballistic filler enclosed in an oxford aramid cloth shell.

1.2 Classification. The body armor trousers shall be of one type in the following sizes as specified (see 6.2):

X-Small	Medium	Large
Small		X-LArge

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8470

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-T-5038	-	Tape, Textile and Webbing, Textile, Reinforcing Nylon
MIL-F-21840	-	Fastener Tapes, Hook and Loop, Synthetic
MIL-L-35078	-	Loads, Units: Preparation of Semiperishable Subsistance Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43842	-	Cloth, Oxford, Aramid
MIL-C-44050	-	Cloth, Ballistic, Aramid
MIL-T-44100	-	Thread, Para-Aramid, Spun, Intermediate Modulus

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case the document shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.3 Materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document.

3.3.1 Cloth, oxford, aramid. The cloth for the outer and inner shell leg and spat covers, fastener straps, and belly band shall be oxford aramid cloth, Olive Green 106, conforming to MIL-C-43842.

3.3.2 Cloth, ballistic, aramid. The cloth for ballistic inserts shall be water-repellent aramid ballistic cloth conforming to type II, class 2 of MIL-C-44050.

3.3.3 Thread. The thread for stitching throughout the trousers shall be para-aramid, spun, intermediate modulus conforming to MIL-T-44100, and shall be Olive Drab S-1, C.A. 66022. Thread size shall be as specified in table II.

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3.3.4 Webbing, textile, nylon. The 2 inch wide nylon webbing for joining ballistic insert panels and attaching the tops of the trouser inserts to the top of the waist, and the 1-1/2 inch wide webbing for suspenders shall be type IV, Olive Drab 7, conforming to MIL-T-5038.

3.3.5 Tape, fastener. The 2 inch wide nylon fastener tapes (hook and pile) used on the waist of the trousers, spats, and leg straps shall be Olive Green 106, type II, class 1 of MIL-F-21840.

3.3.6 Labels. The trousers shall have a combination identification and size label conforming to type VI, class 4 of DDD-L-20.

3.3.7 Buckle. The buckle for the suspender system shall be adjustable with quick-release mechanism, 1/2 inch Fastex #SR-1, or approved equivalent.

3.4 Design. The body armor trousers shall consist of an aramid ballistic filler with an outer aramid oxford fabric shell. The trousers are of a front wrap type with adjusting straps on both legs. The trousers have suspendered tops, two part overlapping knees, attached spats, and attachable belly band.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government. The working patterns shall be identical to the standard patterns which shall not be altered in any way.

3.5.1 Pattern parts. The components of the trousers shall be cut from the materials specified in table I in accordance with pattern parts indicated.

TABLE I. List of pattern parts

Material	Pattern part	Cut parts
Oxford aramid cloth	Upper leg, outer ply	2
	Upper leg, inner ply	2
	Lower leg, outer ply	2
	Lower leg, inner ply	2
	Waist band	2
	Spat, outer	2
	Spat, inner	2
	Spat leg, outer	2
	Spat leg, inner	2
	Leg strap, upper thigh, hook	2
	Leg strap, lower thigh, hook	2
	Leg strap, upper ankle, hook	2
	Leg strap, lower ankle, hook	2
	Leg strap, upper thigh, pile	2
	Leg strap, lower thigh, pile	2
	Leg strap, upper ankle, pile	2
	Leg strap, lower ankle, pile	2

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TABLE I. List of pattern parts (cont'd)

Material	Pattern part	Cut parts
Oxford aramid cloth (cont'd)	Pocket thigh	2
	Spat gusset	2
	Spat, sole bottom	4
Ballistic aramid cloth	Upper leg insert #1	4
	Upper leg insert #2	4
	Upper leg insert #3	4
	Upper leg insert #4	4
	Lower leg insert #1	8
	Lower leg insert #2	8
	Lower leg insert #3	8
	Lower leg insert #4	8
	Spat insert	24
	Spat leg insert	24
	Spat, sole insert	4

3.6 Construction. The construction shall conform in all respects to the requirements specified in table II and herein.

3.6.1 Stitches, seams and stitching. All stitches, seams, and stitching shall conform to FED-STD-751. The type of seam, stitching, and stitches per inch shall be as specified in table II. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used.

3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1 inch except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.6.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching. 1/

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b. Thread breaks or two or more consecutive skipped or run-off stitches noted during end item inspection shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area and continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.1.2 Type 101, 502, 503, 504, and 505 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching. All repairs shall be in accordance with 3.6.1.1.1a and b.

3.6.1.2.1 Optional type 101 holding stitching. Any additional holding stitching to facilitate manufacturing is permissible provided that during ballistic filler assembly the stay stitching is 1/4 ± 1/16 inch from edge of ballistic filler.

3.6.1.3 Bartacks. Unless otherwise specified, bartacks shall be 3/4 inch long with 42 stitches per tack.

3.6.2 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, size and type of thread are met, and at least three or more tying, overlapping or back stitches are used to secure the ends of the stitching.

3.6.3 Thread ends. All thread ends that are visible on the finished trousers shall be trimmed to a length of not more than 1/4 inch.

3.6.4 Lubrication of thread. Except for fabricating the ballistic filler components (see 6.5), there shall be no lubrication of the thread by any means prior to or during sewing (see 4.4.1.1).

3.6.5 Location marks. Marks for showing location of component parts on the outer shells may be done with marking ink or by use of drill holes providing they are located such that the marks or holes are completely and permanently covered by another component part.

3.6.6 Repairs. Repairs such as mends, darns, patches, or splices are not permitted on the trousers.

3.7 Manufacturing operations requirements. The trousers shall be manufactured in accordance with operations specified in table II. The contractor is not required to follow the exact sequence of operations. Any additional basting or holding stitches used to facilitate manufacture are permissible provided the thread does not show on the finished trousers.

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TABLE II.

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting.						
	a. The body armor fabric component shall be cut in strict accordance with patterns furnished which show size, cutting directional lines, placement of components and notches for proper assembling of all parts.						
	b. Ballistic fillers and inserts may be cut in any direction, in strict accordance with patterns furnished.						
	c. All components visible on the exterior of the body armor trousers shall not be drilled for positioning any part except where a component part, when attached, will completely and permanently cover the drill hole.						
	d. Each webbing or fastener tape component shall be cut in the required length as specified or as indicated by patterns ± 1/8 inch. All nylon webbing ends shall be fused.						
2.	Assembly suspenders.						
	a. Slip suspender webbing to be attached to ballistic filler through the end of buckle and fold in half. Stitch across webbing as close to the buckle as possible.	301	SSbf(a)-1	8-10	T-35	T-35	
	b. Position end of longest piece of webbing at an angle, according to marks on patterns, between both layers of folded webbing. Make box-x stitch following	301		8-10	T-35	T-35	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Assemble suspenders. (cont'd)						
	angle of end of inserted webbing. Stitch along both sides of webbing 1/16 to 1/8 inch from edge for 4 to 4-1/4 inches measured from buckle stitching. Repeat with other strap.						
	c. Fold in ends of webbing piece for interlocking suspenders 1/4 to 3/8 inch. Place on webbing strap according to marks on patterns and stitch both ends 1/16 to 1/8 inch from ends. Repeat on other strap.	301	SSb-1	8-10	T-35	T-35	
	NOTE: When stitching second strap be sure to interlock suspender straps by slipping short piece between short piece and long piece on the other strap before stitching.						
3.	Overedge ballistic fillers.						
	a. Each upper leg insert group shall consist of four plies. Each lower leg insert group shall consist of eight plies. Align plies of each group and overedge periphery with 3/8 to 1/2 inch wide stitch gauge.	502, 503, 504, or 505	SSa-1	8-10	T-50	T-50	
	b. Each spat insert group shall consist of eight plies. Each leg spat insert group shall consist of eight plies. Each spat sole insert shall consist of two plies. Align plies of each group and overedge periphery with 3/8 to 1/2 inch wide stitch gauge.	502, 503, 504, or 505	SSa-1	8-10	T-50	T-50	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Assemble ballistic fillers.						
	a. Position all groups of ballistic fillers one on top of another from large to small, evenly aligned and stitch around periphery 1/2 to 3/4 inch from edge, catching all plies. NOTE: Upper leg inserts - 4 groups of 4 plies Lower leg inserts - 4 groups of 8 plies Spat inserts - 3 groups of 8 plies Leg spat inserts - 3 groups of 8 plies Spat sole inserts - 1 group of 2 plies	301	SSbd-1	8-10	T-50	T-50	
	b. Position filler support webbing at top of upper leg inserts per marks on patterns and box stitch in place.	301	LSbj-1	8-10	T-50	T-50	
	c. Position 1-3/4 inch wide front joining webbing to left and right front filler groups at front center seam per marks on patterns and box stitch in place joining fronts.	301	LSbj-1	8-10	T-50	T-50	
	d. Position one suspender strap to each ballistic upper leg, insert according to marks on patterns, making sure not to twist straps, and box stitch in place.	301	LSbj-1	8-10	T-50	T-50	
5.	Make leg straps and thigh pockets.						
	a. With right sides together, stitch leg straps, leaving one end open. Turn and force out corners.	301	SSa-1	8-10	T-35	T-35	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make leg straps and thigh pockets. (cont'd)						
	b. Edge stitch 3/16 to 1/4 inch from edge of seamed sides.	301	SSe-1	8-10	T-35	T-35	
	c. Position hook or pile fastener tape on appropriate leg strap according to marks on patterns. Stitch 1/8 to 1/4 inch from edge on all four sides of fastener tape.	301	LSbj-1	8-10	T-35	T-35	
6.	Assemble tool pocket.						
	a. Overedge periphery of tool pockets.	502, 503, 504 or 505		8-14	T-35	T-35	
	b. Turn top edge of pockets in approximately 1/2 inch and again 1 inch, stitch 1/16 to 1/8 inch from folded edge forming hem.	301	LSbj-1	8-10	T-35	T-35	
	c. Per marks on pattern fold side and bottom extensions to the right side and edge stitch along the fold 1/32 to 1/16 inch from edge.	301	SSb-2	8-10	T-35	T-35	
	d. With right sides together stitch corner seam.	301	SSa-1	8-10	T-35	T-35	
	e. Turn right side out and raise stitch 1/32 to 1/16 inch from edge of corner seam.	301	LSq-2	8-10	T-35	T-35	
	f. Per second marks on pattern fold side and bottom to the inside and edge stitch along the fold.	301	SSb-2	8-10	T-35	T-35	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make upper legs.						
	a. Position label to top of right upper leg inner ply per marks on patterns and stitch 1/8 to 1/4 inch from edge on all four sides.	301	LSbj-1	8-10	T-35	T-35	
	b. Position two 5 inch strips of 2 inch pile fastener tape (abutted along length) to top of right and left upper leg outer plies and one 5 inch strip of 2 inch pile fastener tape towards the bottom of outer plies, per marks on patterns and stitch 1/8 to 1/4 inch from edge on all four sides.	301	LSbj-1	8-10	T-35	T-35	
	c. Position tool pockets to outer plies per marks on patterns and stitch 1/16 to 1/8 inch from edge of turned in seam allowance.	301	SSa-1	8-10	T-35	T-35	
	d. With right sides together join center front seams of upper legs, inner and outer plies respectively.	301	SSa-1	8-10	T-35	T-35	
	e. Overedge front center seams of inner and outer plies respectively.	502, 503, 504, or 505	SSa-1	8-10	T-35	T-35	
	f. Raise stitch 1/16 to 1/8 inch from front center seam edge of inner and outer plies respectively.	301	LSq-2	8-10	T-35	T-35	
	g. With right sides of inner and outer plies of upper legs together, position eight leg straps between the plies in accordance with marks on patterns. The ends of the straps shall be even with the edges of shell material, with pile fastener straps on inseams (pile	301	SSa-1	8-10	T-35	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make upper legs. (cont'd)						
	towards outer ply), and hook fastener straps on outseams (hook towards inner ply). Stitch around periphery except that waist shall be left open to insert fillers.						
	h. Overedge periphery seams and around waist opening.	502, 503, 504, or 505	SSa-1	8-10	T-50	T-50	
	i. Turn right side out and raise stitch 1/16 to 1/8 inch from seams.	301	LSq-2	8-10	T-35	T-35	
8.	Make lower legs.						
	a. With right sides of inner and outer plies of lower legs together, (left and right respectively) position eight leg straps (four to each leg) between plies in accordance with marks on patterns, with ends aligned with seam edges and with fastener tapes as written for upper legs in operation 7.g. Stitch around periphery except than leg bottoms shall be left open to insert filler.	301	SSa-1	8-10	T-35	T-35	
	b. Overedge periphery seams and around bottom openings.	502, 503, 504, or 505	SSa-1	8-10	T-50	T-50	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Make lower legs. (cont'd)						
	c. Turn right side out and raise stitch 1/16 to 1/8 inch from seams.	301	LSq-1	8-10	T-35	T-35	
9.	Insert leg fillers and close shell.						
	a. Insert upper leg filler into upper leg shell, pushing filler to the bottom for a snug fit. Following curve of bottom seams, stitch 1 to 1-1/4 inches from edge across bottoms. Stitch across tops of filler 1 to 1-1/4 inches from edge of filler. Tack all ends.	301	SSe-1	8-10	T-35	T-35	
	b. Insert lower leg filler into lower leg shells, pushing fillers to the tops for a snug fit. Following curve of top seams, stitch 1 to 1-1/4 inches from edge across tops.	301	SSe-1	8-10	T-35	T-35	
	c. Turn seam allowance for waist and leg bottoms in and stitch 1/16 to 1/8 inch from edge.	301	SSc-1	8-10	T-35	T-35	
10.	Attach lower legs to upper legs and bartack.						
	a. Position lower legs over upper legs per marks on on patterns, and stitch in place on both sides with a 1 inch box stitch.	301		8-10	T-35	T-35	
	b. .Bartack each box stitch four times for reinforcement.	Bartack		42 per tack	T-35	T-35	
	c. Bartack each leg strap (centered) 1 to 1-1/4 inches from seam edge catching all plies and filler.	Bartack		42 per tack	T-35	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Make spat shell.						
	a. With right sides together stitch ends of spat gussets together and turn right side out.	301	SSa-1	8-10	T-35	T-35	
	b. Overedge stitch around unseamed edges of spat gussets and around periphery of spat sole	502, 503, 504, or 505		8-14	T-35	T-35	
	c. With right sides out fold spat sole in half, insert ballistic filler, and seam around the outside of the spat sole.	301	SSa-1	8-10	T-35	T-35	
	d. Position spat gussets to spat soles per marks on patterns and stitch in place.	301	SSa-1	8-10	T-35	T-35	
	e. Position a 6 inch strip of 2 inch pile fastener tape to outer spat shell per marks on patterns and stitch 1/8 to 1/4 inch from edge on all four sides.	301	LSbj-1	8-10	T-35	T-35	
	f. With right sides of inner and outer spat together, spat gusset and sole inserted with sole to inner ply, and all edges aligned, stitch around spat leaving tops opened.	301	SSa-1	8-10	T-35	T-35	
	g. Overedge seam and around opening.	502, 503, 504, or 505	SSa-1	8-10	T-35	T-35	
	h. Turn right side out and raise stitch 1/16 to 1/8 inch from seam.	301	LSq-2	8-10	T-35	T-35	

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					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Make spat shell. (cont'd)						
	i. Insert filler in spat, pushing filler to bottom for a snug fit and following curve of spat seam, stitch 1 to 1-1/4 inches from edge around spat bottom catching filler . Tack ends.	301	SSe-1	8-10	T-35	T-35	
12.	Complete spats.						
	a. Overedge periphery of inner and outer leg spat shells.	502, 503, 504, or 505		8-10	T-35	T-35	
	b. Position a 5 inch strip of 2 inch pile fastener tape to outer leg spat shell per marks on patterns, and stitch 1/8 to 1/4 inch from edge on all four sides.	301	LSbj-1	8-10	T-35	T-35	
	c. With right sides of inner and outer leg spat together, spat inserted (pile fastener to pile fastener), attaching webbing inserted (centered) and all edges aligned, stitch bottom and sides closed. Turn right side out.	301	SSa-1	8-10	T-35	T-35	
	d. Insert filler in leg spat, pushing filler to the bottom for a snug fit and stitch across top of filler 1/2 to 3/4 inch from edge of filler catching all plies. Tack ends.	301	SSe-1	8-10	T-35	T-35	
	e. Turn seam allowance for top of leg spat in and stitch 1/16 to 1/8 inch from edge.	301	SSc-1	8-10	T-35	T-35	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Complete spats. (cont'd)						
	f. Turn ends of webbing for straps under and stitch one to leg spat and one to spat per marks on patterns using a 1 inch box stitch.	301	LSbj-1	8-10	T-35	T-35	
	g. Turn opposite ends of spat webbing straps to inside 1/2 inch and positioning a 5 inch strip of 2 inch hook fastener tape over folded edge and stitch on all four sides 1/8 to 1/4 inch from edges.	301	LSbj-1	8-10	T-35	T-35	
13.	Attach spat to trousers.						
	Position spat attachment webbing to outside of trouser legs per marks on patterns and turning webbing under 1/2 inch stitch in place using a box stitch.	301	LSbj-1	8-10	T-35	T-35	
14.	Make waist band.						
	a. With face sides together and edges aligned, stitch around three sides of the belly band plies. Turn face side out and push out corners.	301	SSa-1	8-10	T-35	T-35	
	b. Turn seam allowance of the opened end of belly band in and stitch end closed 1/16 to 1/8 inch from edge. Continue stitching around belly band for edge stitching.	301	SSc-1	8-10	T-35	T-35	
	c. Position two abutted strips of 2 inch hook fastener tape to each end of waist band with ends and side edges aligned and stitch on all four sides of each strip 1/16 to 1/8 inch from edges.	301	-	8-10	T-35	T-35	

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3.8 Finished dimensions. The finished dimensions for the body armor trouser shall conform to table III.

TABLE III. Finished dimensions (in inches)

	X-Small	Small	Medium	Large	X-Large	Tolerance
1/2 Waist 1/	15	16	17	18	19	± 1/2
Inseam 2/	31	31-1/2	32	32-1/2	33	± 3/4

1/ Measure waist (relaxed) from side edge to side edge.

2/ Measure from seam line at base of crotch to bottom of leg.

3.9 Workmanship. The finished body armor trousers shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.2 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the thread lubrication restriction conforms to the requirement specified in 3.6.4.

4.4.2 In-process inspection.

4.4.2.1 In-process examination of cut parts, fillers, and patterns. All cut parts for the shell and filler assemblies, shall be examined during the cutting process to determine that parts are cut properly with respect to size and material directional requirements and that parts containing defects such as a hole, cut, smash, float, loose slub, hard crease, or wrinkles are removed from production. All ballistic fillers shall be examined during assembly of the individual groups to assure that they contain the correct amount of plies. In addition to the above, inspection shall be made of the working patterns to assure that they conform to the Government patterns in all respects. Whenever nonconformance is noted, correction shall be made to the item or items affected. Parts which cannot be corrected shall be removed from production.

4.4.2.2 In-process size examination for filler groups. The ballistic filler groups after edge stitching shall be examined for the defects listed in table IV. The lot size shall be expressed in terms of left or right front or back

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filler groups, as applicable. The sample unit shall be one left or right front or back filler group, as applicable. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.0 for major defects and 4.0 for total (major and minor combined).

TABLE IV. In-process size defects

Examine	Defect	Classification Major	Classification Minor
Size of individual ballistic filler	a. Smaller than cutting pattern		
	(1) 3/16 inch to 3/8 inch at any point around periphery 1/ 2/		X
	(2) More than 3/8 inch at any point around the periphery 1/ 2/	X	
	(3) 1/8 inch up to 3/16 inch around entire periphery 2/		X
	(4) More than 3/16 inch around entire periphery 2/	X	
	b. Larger than cutting pattern by 3/8 inch or more at any point 1/ 2/		X

1/ To be scored when condition exists for a length of more than 2 inches or if condition exists in several areas with an accumulated distance of 4 inches or more.

2/ The individual component, as applicable, shall be examined with the applicable cutting pattern centered on the filler component.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table V. The lot size shall be expressed in units of trousers. The sample unit shall be one pair of trousers. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 15.0 for total (major and minor combined) defects.

TABLE V. End item visual defects

Examine	Defect	Classification Major	Classification Minor
Cloth	a. Any hole (except drill hole), cut, or tear.	X	
	b. Any exposed drill hole.		X
	c. Any abrasion marks, broken or missing yarns, or multiple floats.	X	
	d. Any mend, darn, or patch.	X	
	e. Parts not cut correctly with respect to material lines.	X	
	f. Needle chews.	X	

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TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Webbing	a. Any hole, cut, or frayed edge.	X	
	b. Needle chews:		
	(1) Up to 1/8 inch in length.		X
	(2) 1/8 inch or more in length.	X	
	c. Not continuous, i./e. spliced.	X	
	d. End not fused as required.		X
Fastener tape	Hook tape crushed affecting closure.	X	
Seams and stitching:			
Open seams	a. Up to and including 1/2 inch.		X
	b. More than 1/2 inch.	X	
NOTE:	A seam shall be classified as an open seam when one or more stitches joining a seam are broken, or when two or more consecutive skipped or runoff stitches occur.		
Raw edges (except where required or otherwise classified herein)	More than 1/2 inch when securely caught in stitching.		X
NOTE:	Raw edges not securely caught in stitching shall be classified as open seams.		
Seam and stitch type	a. Wrong seam or stitch type.	X	
	b. Seam pleated or badly puckered, clearly noticeable.		X
Stitch tension	a. Tension loose, resulting in loose bobbin or top thread.		X
	b. Tension tight, resulting in puckering of material.		X
Stitches per inch	a. One to two stitches less than minimum specified 1/		X
	b. Three or more stitches less than minimum specified 1/	X	
	c. Two or more stitches in excess of maximum specified 1/		X

1/ Defects to be scored only when condition exists for a length of 3 inches or more in several areas with an accumulated distance of 5 inches or more. Applicable to individual seams.

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TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Seams and stitching: (cont'd)			
NOTE:	Variation in the number of stitches per inch caused by operator speeding up the machine and pulling the material in order to sew over heavy places, or in turning corners shall be classified as follows: (a) Within the minor defect classification - No defect (b) Within the major defect classification - Minor defect		
Thread breaks, or skipped stitches (on type 301 stitching)	Overstitched less than 1 inch in each direction beyond defective stitching area		X
NOTE:	On all types of stitching, thread break or two or more consecutive skipped stitches not overstitched shall be classified as open seam.		
Stitching end (on type 301 stitching)	End of stitching not secured as specified (except when caught in other stitching or turned under in a hem)		X
Runoffs	a. On joining or closing seams - use open seam classification. b. Edge or raised stitching, not resulting in any open seam, more than 1/2 inch in length (when resulting in an open seam use open seam classification.)		X
Rows of stitching	Any row missing (unless otherwise classified herein).	X	
Bartacks	a. Stitching loose, incomplete, or broken.		X
	b. Any bartack omitted.	X	

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TABLE V. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Assembly - general	a. Any component caught in any unrelated operations of stitching, affecting appearance or function (not otherwise classified herein.)	X	
	b. Any component part omitted (not otherwise classified herein) or not attached as specified.	X	
	c. Any row of stitching securing ballistic filler to inner shell omitted.	X	
	d. Any unauthorized repair.	X	
Label (size and identification)	a. Missing, incorrect, or illegible.	X	
	b. Misplaced.		X
Cleanness	a. Any spot or stain clearly noticeable.		X
	b. Thread ends not trimmed as specified.		X
Color of components	Color of any component not as specified.		X

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified in table III. Any dimension deviating from the specified requirement shall be classified as a defect. The lot size shall be expressed in units of trousers. The sample unit shall be one pair of trousers. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.

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Examine	Defect
Workmanship	Inadequate application of components, such as: incomplete closure of flap, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number of trousers per container is more or less than required.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size location, sequence, or method of application.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each pair of trousers shall be laid flat, front down with all fasteners closed. The spats shall be folded up onto the back of the lower leg and the waist and waistband shall be folded down over the back of the hip area. The trousers shall then be folded in half at the center seam bringing the two side seams together. The folded trousers shall measure approximately 38 inches in length and 15 inches in width.

5.1.2 Commercial. Trousers shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

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5.2.1 Level A packing. One pair of trousers, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style FTC, grade V2s of PPP-B-636. Trousers shall be completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. The inside dimensions of the container shall approximate 38 inches in length, 15-1/2 inches in width, and 3-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. One pair of trousers, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style FTC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Trousers shall be completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. The appropriate inside dimensions of the shipping container shall be as specified in 5.2.1. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Trousers, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), trousers packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be number 3 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandataory.)

6.1 Intended use. The body armor is intended for use for protection against high velocity fragments associated with Unexploded Ordnance (UXO) and Improvised Explosive Devices (IED).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Size required (see 1.2).
c. Issue of DODIS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article sample is required (see 3.2, 4.3, and 6.4).
e. Selection of the applicable level of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.5 Cooling of needle. In order to keep the needle cool when overedging the ballistic fillers, it is recommended that either the machines be equipped with a compressed air blower directed at the needle or by affixing a felt pad saturated with silicone oil in a position so that the thread will rub against it during sewing. The use of chromium plated needles have been found to be advantageous in that they do not generate as much heat or deflect as much during use as regular needles. Recommended maximum speed for the overedge machine is 1000 RPM.

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6.6 Subject term (key word) listing.

Body armor
Protective

6.7 Changes from previous issue. Marginals notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Air Force - 99

Review activities:
Army - MD
Air Force - 82
DLA - CT

Preparing activity:
Army - GL

Project No. 8470-0126

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-0044266A(GL)	Body Armor, Trousers, Explosive Ordnance Disposal (EOD) System

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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